

EMCH380 – Project Management Work Breakdown Structure

Slides are developed from instructor's notes and other references.

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Detailed Planning

Work Breakdown Structure

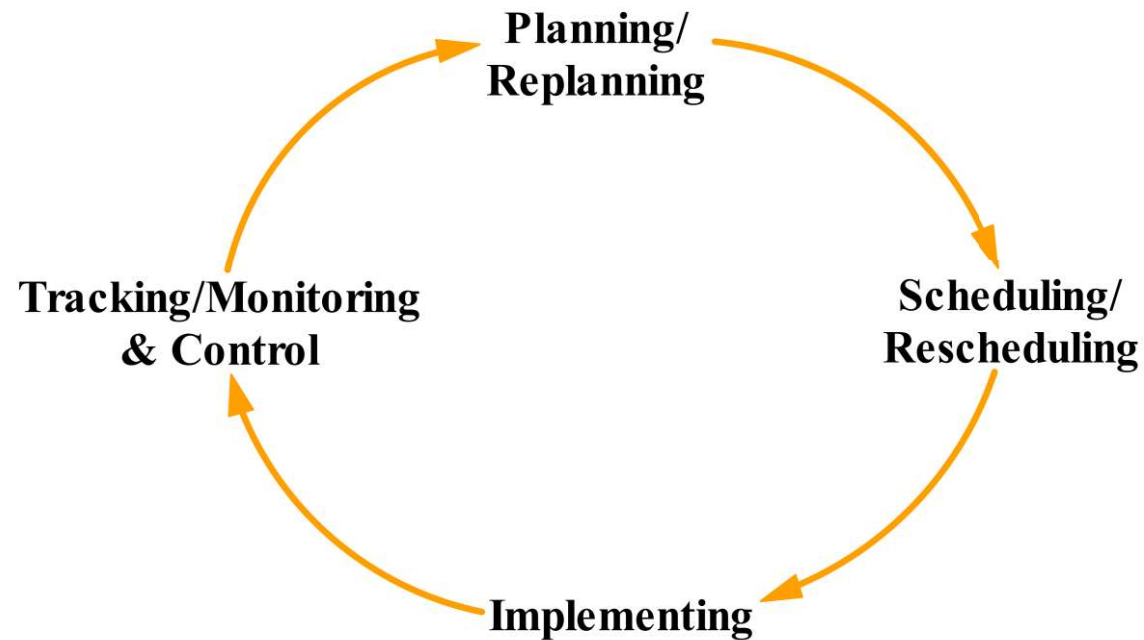
PROJECT PLANNING LIFECYCLE PHASES (BY KERZNER)

- Conceptualization
- Feasibility
- Preliminary Planning
- **Detailed Planning**
- Execution
- Testing and Commissioning

DETAILED PLANNING

- Involves the development of detailed project schedule and budget
 - Development of a Work Breakdown Structure (WBS)
 - Networking activities in WBS
 - Generating Detailed schedule of activities in WBS
 - Developing detailed cost estimates of activities in WBS
- Detailed plan will serve as the basis for monitoring project progress and controlling deviations from plan.

DETAILED PLANNING & MONITORING



WBS

Work Breakdown Structure

WORK BREAKDOWN STRUCTURE (WBS)

- The WBS is the breaking down of the SOW into smaller elements that are:
 - Manageable, in that specific authority and responsibility can be assigned
 - Independent, or with minimum interfacing with and dependence on other ongoing elements
 - Integratable so that the total package can be seen
 - Measurable in terms of progress.

COMMON WBS

- Most common is the six-level indented structure:

	Level	Description
Managerial Levels	1	Total Program
	2	Project
	3	Task
Technical Levels	4	Subtask
	5	Work Package
	6	Level of Effort

COMMON WBS (CONT'D)

- The upper 3 levels of the WBS are normally specified by the customer is part of an RFP/RFQ as the summary levels for reporting purposes
- The lower levels are generated by the contractor for in-house control
- The summation of all elements in one level must be the sum of all work in the next level

COMMON WBS (CONT'D)

- Level 1 is generally used for authorization and release of all work
- Budgets are prepared at level 2
- Schedules are prepared at level 3
- Effort required by departments or sections should be defined at level 4
- The level at which the work is managed is called the package level (level 5)

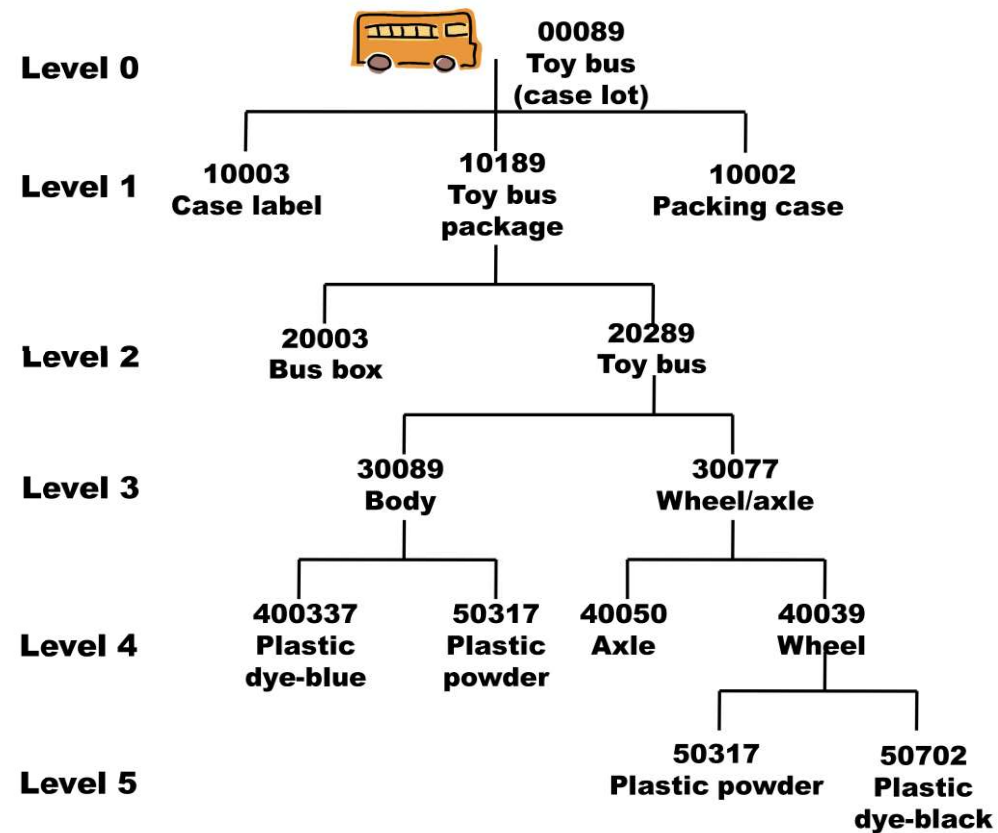
COMMON WBS (CONT'D)

- PM normally manage at the top 3 levels of a WBS and provide status report to management at these levels also
- In some companies the top 3 levels are standardized across projects. Only the lower 3 levels are project specific.
- This is only feasible for companies with great deal of similarity among projects
- Work packages are supervised and performed by line managers with status reporting provided to the project manager at higher level of the WBS

WBS EXAMPLE – CONSTRUCTION

<u>Program:</u> New Plant Construction and Start-up	01-00-00
<u>Project 1:</u> Analytical Study	01-01-00
Task 1: Marketing/Production Study	01-01-01
Task 2: Cost Effectiveness Analysis	01-01-02
<u>Project 2:</u> Design and Layout	01-02-00
Task 1: Product Processing Sketches	01-02-01
Task 2: Product Processing Blueprints	01-02-02
<u>Project 3:</u> Installation	01-03-00
Task 1: Fabrication	01-03-01
Task 2: Setup	01-03-02
Task 3: Testing and Run	01-03-03
<u>Project 4:</u> Program Support	01-04-01
Task 1: Management	01-04-01
Task 2: Purchasing Raw Materials	01-04-02

WBS EXAMPLE – MANUFACTURING



CRITERIA FOR DEVELOPING A WBS

- The WBS and work description should be easy to understand.
- All schedules should follow the WBS.
- No attempt should be made to subdivide work arbitrarily to the lowest possible level.

CRITERIA FOR DEVELOPING A WBS (CONT'D)

II

- Lowest level of work should not end up having a ridiculous cost in comparison to other efforts.
- Maintain flexibility in the WBS because scope of effort can change during a program.
- Most WBS elements (at the lowest control level) range from 0.5% to 2.5% of total project budget.

FACTORS INFLUENCING THE LEVEL OF DETAIL

II

- Overall project duration
- Past performance of teams performing task
- Complexity of task
- Level of risk associated with task
- Dependency of task's completion to follow-on tasks
- Level of visibility task receives
- Rules of thumb determining level of detail
 - duration between 1 to 2 weeks
 - decrease detail if team consistently on time
 - increase detail on key tasks
 - increase detail on highly cross functional tasks

CREATING A WORK BREAKDOWN STRUCTURE

II

- Techniques
 - Brainstorm / Storyboard
 - Top-Down
 - Bottom-Up
- Determine differences between template and your project
- Use action verb followed by a subject
 - Perform QFD (Quality function deployment is a method to transform qualitative user demands into quantitative parameters)
 - Benchmark Competition
- Use notes to define task in more detail

CREATING A WORK BREAKDOWN STRUCTURE (CONT'D)

II

- Use template or previous project as a starting point whenever possible
- Identify phases of the project
- Decompose each phase into major elements
- Decompose each major element into tasks
- Decompose each task into activities (if required)
 - Level of decomposition should reflect amount of information required to manage the project
 - Data is expensive
 - Don't choke yourself with detail

CREATING A WORK BREAKDOWN STRUCTURE (CONT'D)

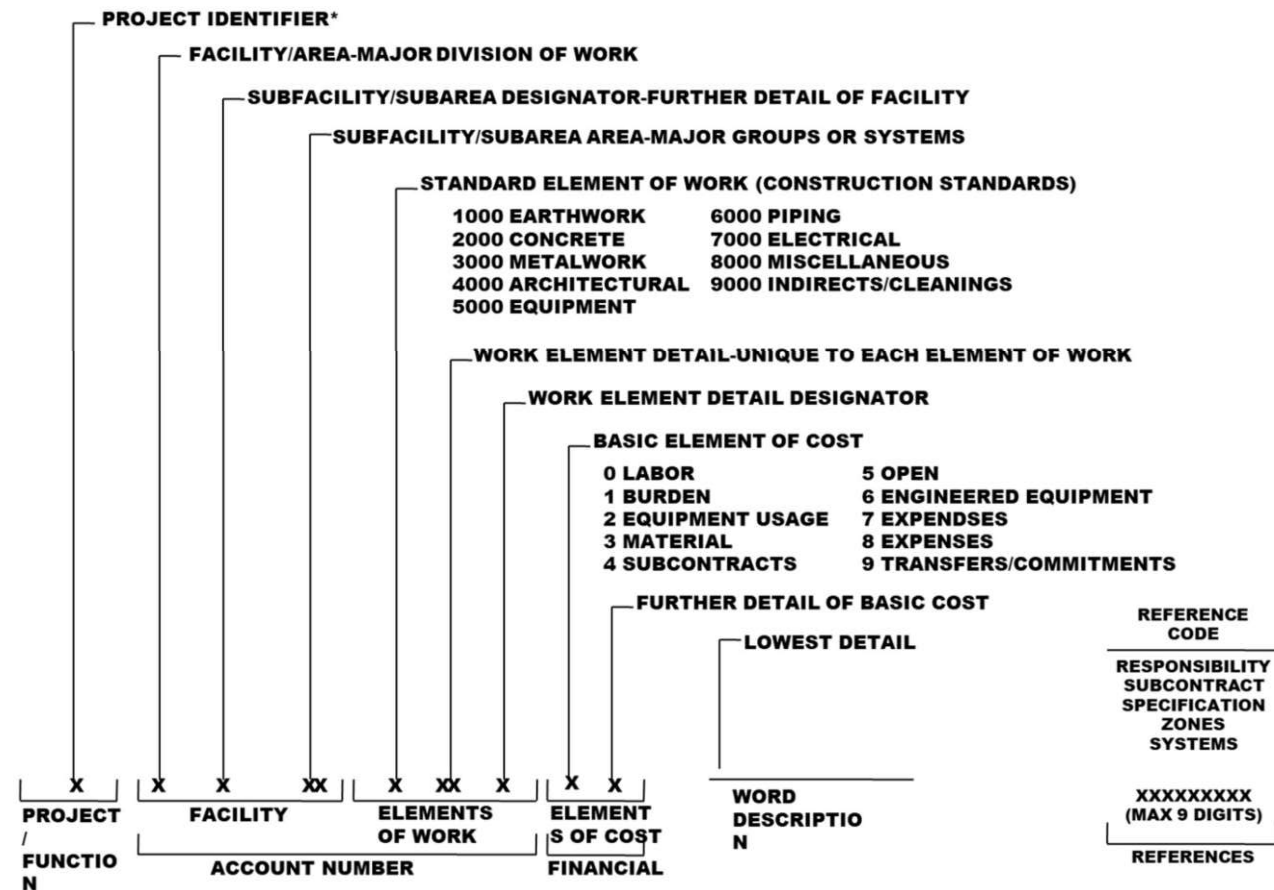
II

- Areas of Responsibility
- Categories of Work as Distinguished by Craft, Equipment, or Materials Requirements
- Location of the Work Within the Project
- Contractor's Breakdown Used for Estimating
- Owner's Breakdown for Bidding or Payment

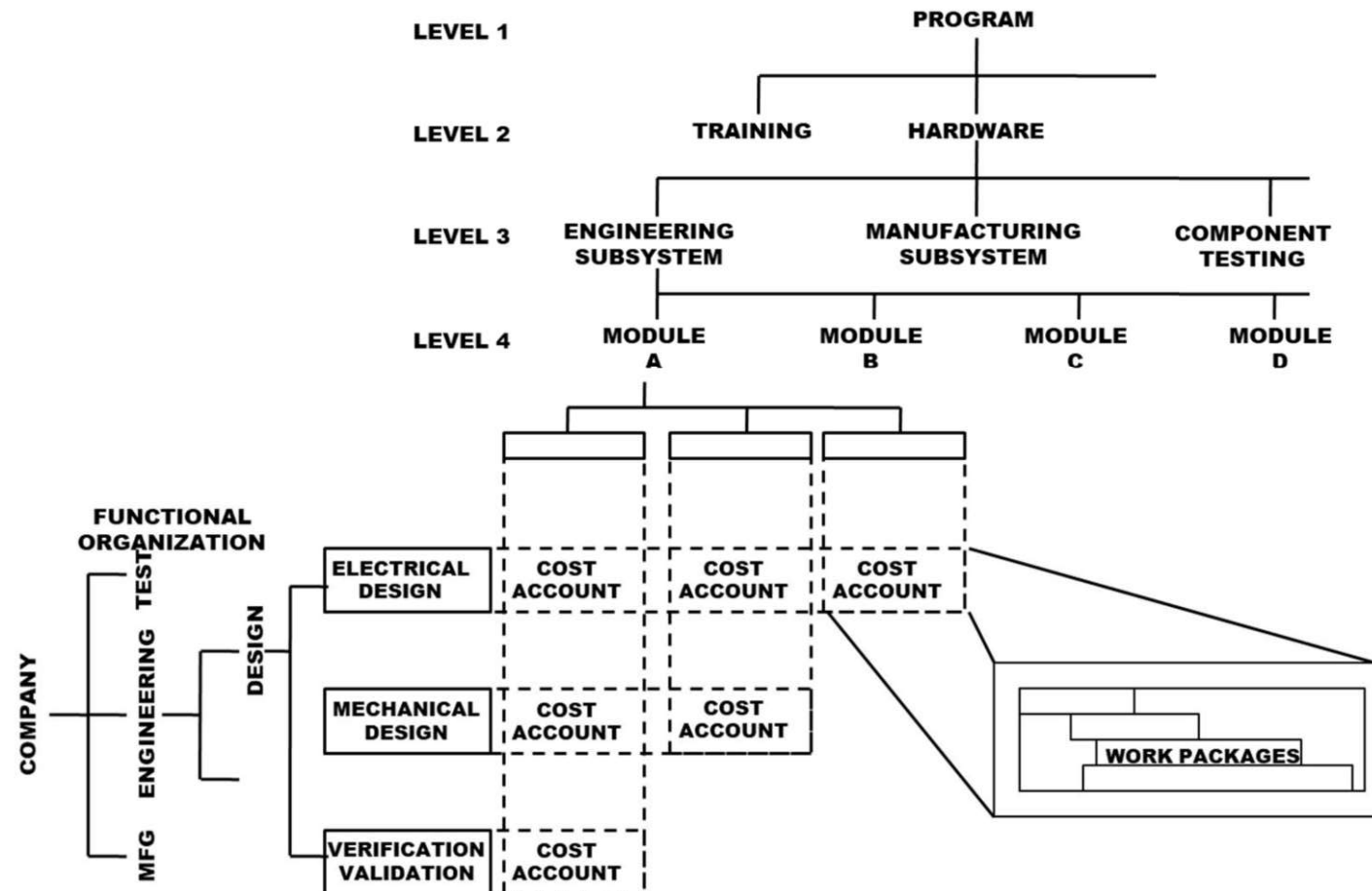
WHY PREPARE A WBS?

- WBS Provides the Basis for:
 - Network scheduling
 - Costing
 - Risk analysis
 - Coordination of objectives
 - Control (including contract administration)

WBS FOR CONSTRUCTION PROJECTS



COST ACCOUNT INTERSECTION



CHARACTERISTICS OF WORK PACKAGES (LEVEL 5)

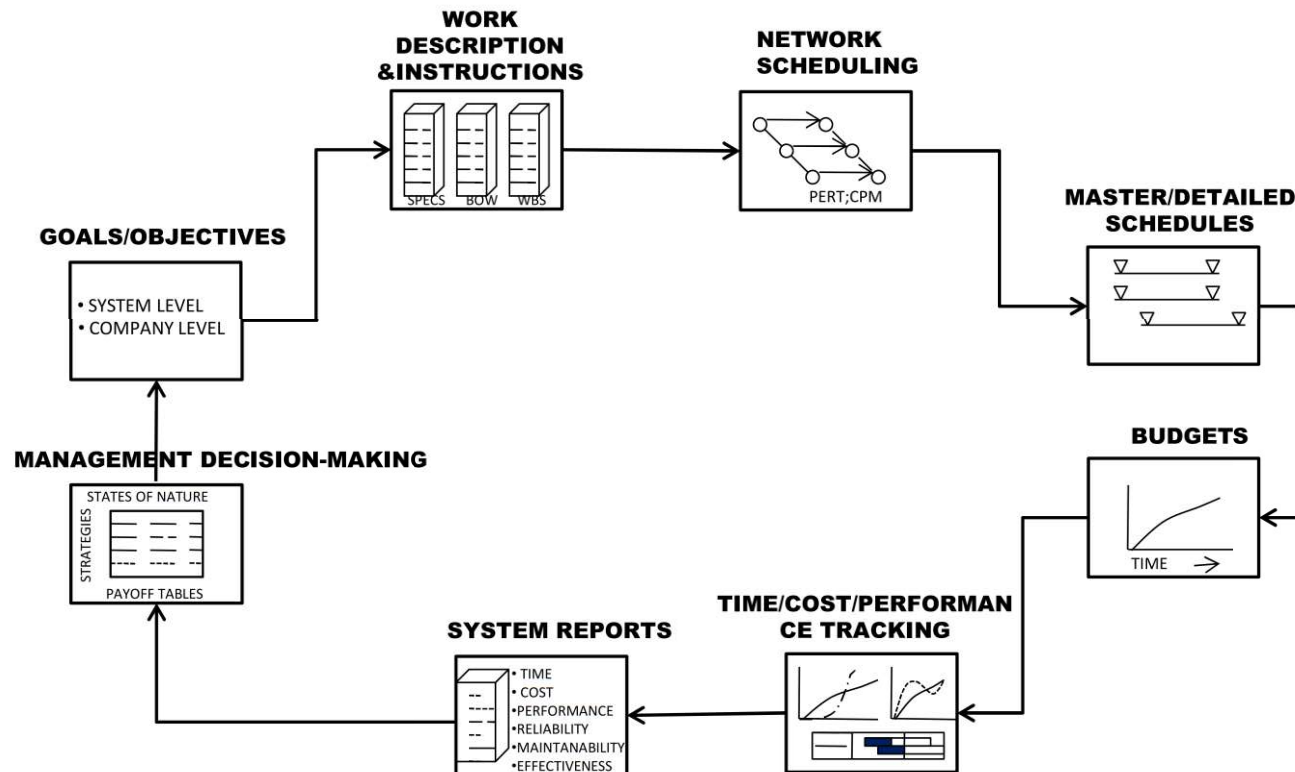
- Have clearly defined start & end date that are representative of physical accomplishment
- Represents units of work at the level where the work is performed
- Specifies a budget in terms of dollars, man-hours, or other measurable units.

Summary

Work Breakdown Structure

PROJECT PLANNING & CONTROL SYSTEM

III



IMPORTANCE OF WBS – SUMMARY

- WBS provides a common framework from which:
 - The total project can be described as the summation of subdivided elements
 - Planning can be performed
 - Costs and budgets can be established
 - Time, cost and performance can be tracked

IMPORTANCE OF WBS – SUMMARY

- Objectives can be linked to company resources in a logical manner
- Schedules and status reporting procedures can be established
- Network construction and control planning can be initiated
- The responsibility assignments for each element can be established.

WBS FOR CONTROL & EVALUATION

